

current as it is equipped with high-current relays operating under 250V, 10A AC or 30V, 10A DC. It has a standard interface that can be controlled

directly by the microcontroller.

This module is optically isolated from high voltages for safety and has four SPDT relays.

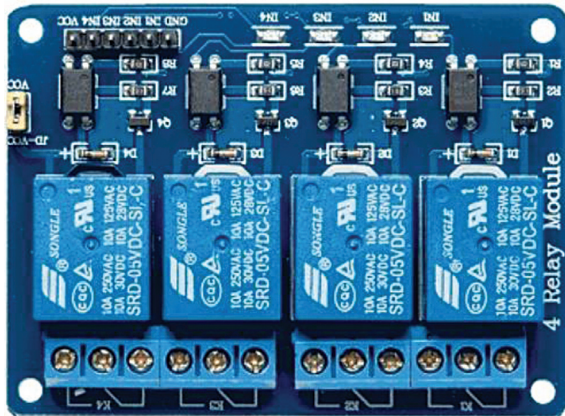


Fig. 3: The 4-channel relay board

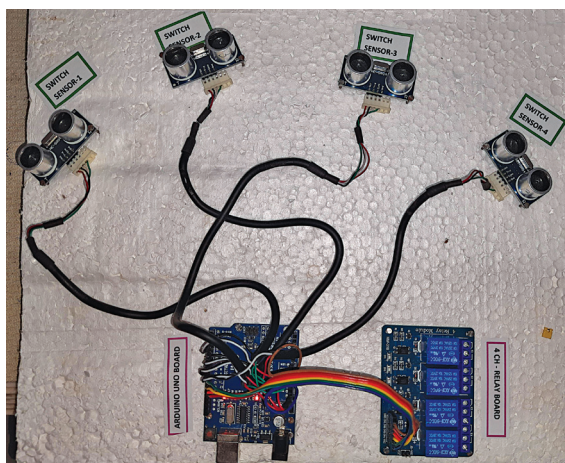


Fig. 4: Author's prototype

Software

The software program *fourchan.ino* is loaded into the internal memory of Arduino Uno. The program is simple and easy to understand. Comments are given at the end of each command line.

The following need to be defined in the code first:

- Echo and trigger pins of the ultrasonic sound sensor
- Four relays' pins
- Intermediate variables for relay operation
- Variables for all the four sensors

Next, a void setup () function should be created. The corresponding pins should be initialised as inputs and outputs. Then the serial communication should be initiated. The serial communication has nothing to do with the project, it is used for just analysing the program on a monitor.

EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

Board1 - Arduino Uno

Miscellaneous:

SEN1-SEN4 - HC - SR04 sensor

RM1 - 5V, 4-channel 1CO relay module

LOAD1-LOAD4 - 230V AC appliances
- Jumper wires

The loop() function allows the program to change and respond. It is used to actively control the Arduino Uno board. Activate the Echo sensor. Calculate the distance and perform the switching action on/off. Pass on the switch condition to the relay. Write the above code for all the other three sensors and relays as well.

Construction and testing

The circuit requires a 12V, 1A DC power supply. Connect the module as shown in the circuit diagram. Maintain some distance between the four Echo sensors so that their operation does not affect the adjacent sensors.

A PVC switchboard box can be used to enclose the Arduino Uno and the relays board. Connect the AC appliances in accordance with the specifications of the relays. The connection of phase (L) and neutral (N) wires is shown in the circuit diagram. The author's prototype is shown in Fig. 4.

Note. The Arduino Uno board and HC-SR04 sensors are highly sensitive, so handle carefully. You need to bring your hand near the sensor slowly and remove it slowly.

Caution. Take proper precaution while handling AC mains. Novice experimenters are advised to consult an expert before making AC mains connections with the relays. **EFY**

Arduino Uno Pin Connections with Components

Table 2

Arduino Uno pins	Components	HC-SR04 sensors
5V pin	• Connect to Vcc pins of all four HC-SR04 sensors • Connect to Vcc pin of relay board	
Pin 2	HC-SR04 -1 Trigger pin	SEN1
Pin 3	HC-SR04 -1 Echo pin	
Pin 4	HC-SR04 -2 Trigger pin	SEN2
Pin 5	HC-SR04 -2 Echo pin	
Pin 6	HC-SR04 -3 Trigger pin	SEN3
Pin 7	HC-SR04 -3 Echo pin	
Pin 8	HC-SR04 -4 Trigger pin	SEN4
Pin 9	HC-SR04 -5 Echo pin	
Pin 13	IN 1 of relay board	
Pin 12	IN 2 of relay board	
Pin 11	IN 3 of relay board	
Pin 10	IN 4 of relay board	
GND	• Connect to GND pins of all HC-SR04 • Connect GND pin of relay board	



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